

careful about, but sometimes they fail in keeping it simple and easy to understand from the consumer point of view. The product would not be successful if it is not implemented in a way the customers find useful and simple.

Another major area where innovators fail is pricing. Countries like India have a price-sensitive market, and unless the product has an affordable price point, and its design blends in with the ecosystem of the country, selling the product could be difficult.

Safety and security concerns

It is important to understand standard safety, security and regulatory requirements, and make sure that the end product satisfies this criterion. Engineers in India often fail to think through and probably do not know or do not even consider that.

Mashelkar said, "Somehow, in India, cost is the only consideration and not safety." If the product is not designed for safety and security, it cannot be certified, and unless it has relevant certifications, it cannot be sold in international markets.

When developers reach this stage, the only solution is to perform reverse engineering and find out what went wrong in the design, to find out where they breached the specification and safety standards and correct those. This requires time and money, further delaying the time-to-market and increasing the cost of the product. Hence, it is imperative to take care of the level of safety, security and reliability during the design stage itself.

Getting certified

Every field in electronics has a set of international standards. But of

“Engineers in India are now open to changes and new ideas and are striving to think globally. If they are provided with the right education and skills, we can make India and Made-in-India brand stronger and take it international.”

— Ian Hennell, the US operations director, LDRA, the USA



course, many countries have their own specific set of standards derived from international standards. This is because products developed in a foreign place would not always fit into every economy or lifestyle.

Some product development companies are going for international certifications and their variations with consultants coming in from countries like Germany, Italy or the USA. But this is very expensive. It is important to build that expertise with as much local resources as possible because in India, price matters.

Automation of software verification, standards compliance, certification and regulatory support are now available through software tool suites provided by LDRA, making the process easier and cheaper.

Developing a favourable ecosystem

Several big brands only own the brand name. Most components and sub-systems that go inside the final product are from different vendors. When setting up base in India and working in a supply-chain system, we need to have vendors who can make these available locally.

In today's scenario, India lacks this ecosystem. The only option for such firms is to import the sub-systems. This is not profitable for the brands as they only have assembly units here. Due to this, they prefer to not set base in India. So while we say that big players are not buying

anything from India for their systems, when these players look at India, they do not find vendors who meet their criteria.

In order to successfully establish a manufacturing unit, vendor base has to be developed and an ecosystem has to be built around it over the years, and this ecosystem should have the capability to become the right vendor for that industry. If we work on increasing the capabilities of indigenous companies and creating a good supply-chain model in India, products would not only sell in international markets but also become a part of major systems and programmes.

While coming up with schemes, the government is focussing a lot on skill building. Electronics associations and other technology players have been actively supporting this initiative. Our professional education system has also been encouraging students with college-level technology business incubation wings. Funding and other help are provided to assist students so they can develop their idea into a product and take it to the market. But we still have a long way to go in moving from an examination-oriented system to an application-oriented one.

Indian product developers have come a long way from avoiding risks, playing safe and preferring comfort zones to being adventurous, passionate and not afraid of failure. We already have the talent, capital, ideas and skill-sets. What Indian innovators and product developer companies require is a systematic process of innovation. Having well-structured processes should improve the rate of conversion of an idea into a successful product in the market. ●

“It is high time that people realise that it is not only about designing a fancy product; equally or more important is to make it acceptable to a market from various angles.”

— Shinto Joseph, operations and sales director, LDRA India

